Reg. No. | 9 | 5 | 1 | 7 | | | | | | | | | |

Question Paper Code : 19306

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2022

Third Semester

B.E. – Computer Science and Engineering

19CS301 – DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

BTL, CO

R, CO1 1. What is asymptotic notation?

A, CO1 2. The time complexity used for inserting a node in a priority queue on the basis of key is _____. Justify your answer.

A) O(n) B) O(n^2) C) O(nlogn) D) O(logn)

R, CO2 3. Give any four applications of stack.

A, CO2 4. A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)? Justify your answer.

A) Both operations can be performed in O(1) time

B) At most, one operation can be performed in O(1) time, but the worst-case time for the other operation will be Ω(n)

C) The worst-case time complexity for both operations will ne Ω(n^2)

D) The worst-case time complexity for both operations will be Ω(log n)

R, CO3 5. Give an example for Expression trees.

A, CO3 6. The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)? Justify your answer.

A) 2 B) 3 C) 4 D) 5

U, CO4 7. Mention the difference between Hashing & Extendable Hashing with an example.

U, CO4 8. Double hashing is one of the best methods available for open addressing. Justify your answer.

A) True B) False

R, CO5 9. Define biconnected graph.

U, CO5 10. What is the number of edges present in a complete graph having n vertices? Justify your answer.

A) (n*(n+1))/2 B) (n*(n-1))/2

C) n D) Information given is insufficient

PART B – (5 × 16 = 80 Marks)

U, CO1 11. a) i. Explain the Sequential and Binary search methods. Compare their performances with an example. (10 Marks)

U, CO1 11. a) ii. Define Big-oh notation. Explain the properties of Big-oh notation. (6 Marks)

OR

U, CO1 11. b) i. What is recurrence equation? Explain the various methods of solving recurrence equation. Explain how to solve recurrence equation with an example. (10 Marks)

U, CO1 11. b) ii. Write short notes on analysis of an algorithm. (6 Marks)

U, CO2 12. a) What are advantages of circular linked list? Write down the various primitive operations performed on circular linked list. Write the algorithm for insertion, deletion of element and searching for an element in circular linked list. (16 Marks)

OR

U, CO2 12. b) Write algorithm for push and pop of a stack. Explain how stack is used for evaluation of expression. Write appropriate algorithm and describe the evaluation of an expression step by step with a suitable example. (16 Marks)

U, CO3 13. a) i. Write algorithm for the following operations in Binary search tree: A) Insert B) Delete (8 Marks)

A, CO3 13. a) ii. Build an AVL tree from the sequence of integers given below: 6, 7, 8, 12, 15, 17, 9 and 10. Show the resultant AVL tree after every insertion. (8 Marks)

OR

U, CO3 13. b) i. What is splay tree? Discuss briefly the various rotations in a splay tree with an example. (8 Marks)

A, CO3 13. b) ii. Construct an expression tree for the following expression *(a+b*c) + ((d*e+f)*g)* and write the three traversal expression for the tree constructed. (8 Marks)

U, CO4 14. a) i. Describe the different hashing functions with an example. (8 Marks)

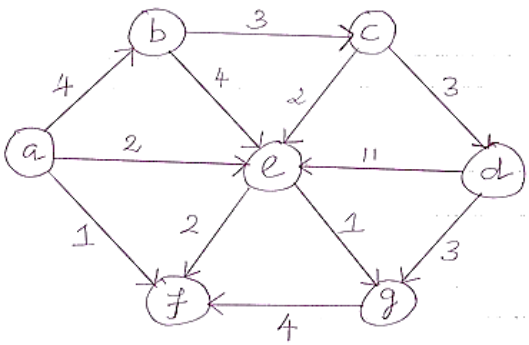
U, CO4 14. a) ii. Explain the common collision resolution strategies in open address hashing. (8 Marks)

OR

U, CO4 14. b) i. Explain the operation of union-by-height with an example. (8 Marks)

U, CO4 14. b) ii. Discuss the disjoint set operation: find using path compression with a suitable example. (8 Marks)

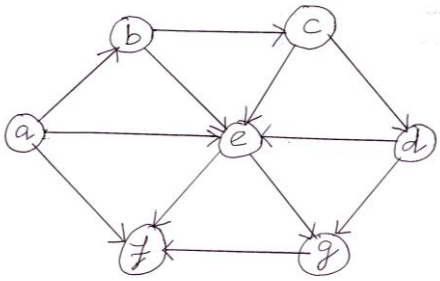
A, CO5 15. a) Explain Dijistra's algorithm and find the shortest path from vertex a to all other vertices in a graph. Write the algorithm and analysis the efficiency.



(16 Marks)

OR

A, CO5 15. b) What is Topological sort? Write down the pseudo code to perform topological sort and apply the same to the following graph. Analysis the efficiency of the algorithm.



(16 Marks)